# Jan Jędrzejewski

Junior ML / AI Engineer · shipping models from idea to production

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Computer Science engineer (Warsaw University of Technology), currently pursuing an MSc with an AI specialization. I combine solid backend and database experience with building ML models deployed as working applications — from data, through training, to deployment. I co-develop production REST APIs in Python, and in my own projects I take models end-to-end (XGBoost, PyTorch, computer vision) all the way to containerization and a public demo. I am looking for my first ML / AI Engineer role where I can grow my experience building production models within a team.

## CORE SKILLS

- **Machine Learning / AI:** PyTorch, TensorFlow, XGBoost, scikit-learn, computer vision (YOLO), predictive modeling
- **Backend:** Python, FastAPI, REST APIs, OOP, API integrations
- **Data & databases:** SQL, PL/SQL, Oracle (EBS / APEX), PostgreSQL / PostGIS, pandas, NumPy, relational design, ERD/UML
- **DevOps & cloud:** Docker, GCP, Firebase, Azure, Git, CI/CD, Jira, Agile
- **Other:** C#, Java, JavaScript, React, Angular

## EXPERIENCE

**Backend Developer (Python) / Co-founder** · Social Surfer *Sep 2024 – Present*

- Designed and maintain the production database and cloud infrastructure for a mobile app: 1,300 users, 32k requests/day, 99.9% uptime.
- Built the Python REST API powering the entire application — 68 endpoints.
- Led the Azure → Google Cloud / Firebase migration, cutting hosting costs by 25%.

**Junior PL/SQL Developer** · Dahliamatic Sp. z o.o. *Jul 2023 – Present*

- Maintain Oracle E-Business Suite for 7 clients — ~80 tickets/month with no critical downtime.
- Built 40+ PL/SQL packages and procedures automating document workflows, reducing processing time.
- Delivered 70+ EBS customizations: reports, forms, and integrations with external systems.

**AI-CCTV — anomaly detection in surveillance footage** · Research project (WUT) *2022 – 2023*

- Recorded and annotated a dataset of 1,500 clips across 13 situation categories.
- Trained models to detect and categorize situations and anomalies — 86% accuracy.

## PROJECTS

**Car Price Estimator — web app for used-car valuation** XGBoost · FastAPI · React · **Live demo** · **Code**

- Valuation model trained on 180k listings, mean error ±7.36% on the test set; served through a FastAPI endpoint with a web form.

**F1 Race Predictor — race-result forecasting** Monte Carlo · FastF1 · FastAPI · React · **Live demo** · **Code**

- Monte Carlo simulation computing finishing-position probability distributions from official timing data (FastF1); FastAPI backend with SQLite cache and a bilingual PL/EN interface.

**Traffic Insights — traffic-volume analysis from public cameras** YOLO · FastAPI · PostgreSQL · Streamlit · Docker · **Code**

- Multi-service system (collector, worker, database, API, dashboard) counting vehicles from public road cameras via object detection; continuous data collection and traffic-pattern visualization. Privacy-by-design — only aggregated counts are stored.

**Route Finder — road-route planning** PostGIS · pgRouting · Leaflett · **Code**

- Optimal route planning over a complete regional road network (Masovia, Poland); routing handled by pgRouting procedures in a spatial database, with results shown on an interactive map.

## EDUCATION & ADDITIONAL INFORMATION

**Warsaw University of Technology** · Faculty of Electronics and Information Technology *2020 – Present*

- MSc in Computer Science, AI specialization (in progress); Engineer's degree (BEng) obtained February 2025.
- Engineering thesis: AI models for radar object recognition — 97% accuracy.

**Languages:** Polish — native · English — B2 (professional working proficiency) · German — basic · French — basic

**Achievements:** Kangaroo Math Competition — laureate · National Physics Olympiad — laureate

**Interests:** volleyball team captain · 300+ km cycling trips · go-kart building (local community project)